

**ADMINISTRATIVE INFORMATION****FMI No: 25343****FMI Title: LightSpeed RT M4 Upgrade**

FMI TYPE	CHARGE CODES	IMPLEMENTATION DATES
☐ Mandatory Safety	GEMS - AM:	Issue Date:
☒ Mandatory	GEMS - E:	
☐ Mandatory on Request	GEMS - A:	Close Date:

☒ Staged FMI	Customer List provided by FMI Administration. If you identify additional units within effectivity range, but they are not shown on the customer list, contact FMI Administration in your Pole.
☐ Parts	and any special tools are furnished automatically. To address shortages and ordering exceptions, contact FMI Administration in your Pole.
☒ Parts Ordering:	GEMS - Americas, Contact your Region logistics coordinator GEMS - Europe, Order FMI kits from GPO/EPL Buc GEMS - Asia, Contact your region logistics coordinator
☐ No Parts	are required to complete this FMI.
☐ Batch with	_____
	To minimize costs, combine implementation of this FMI with the FMI's listed for all customers where the effectivity is the same (Please see your customer lists).
☐ Non-Staged	No Customer List is provided. Each service location must identify affected units, and place part orders as indicated in the Special Instructions.

Technical Support		
GEMS - Americas	GEMS - Europe	GEMS - Asia
GE Medical Systems Insite Center (414) 524-5300 800-321-7973 Milwaukee, WI 53201, USA	GEMS - E / ESC BP34 F 78533 BUC CEDEX, FRANCE Tel: (33 1) 39 20 00 07 or ESC Fax: (33 1) 30 70 99 70	Asia Support Center Phone: 81-426-56-0033 Fax: 81-426-48-2905

Special Instructions:**Unit Tracked: 2374681-2****FMI Instructions Part No.: 5112555-100****Do NOT leave unsecured on site**

FMI Completion Certification Report

Mail To

GEMS - Americas

GEMS - Europe

G.E. Medical Systems
Attn: FMI Admin. W-523
P.O. Box 414
Milwaukee, WI 53201

GEMS - Europe
FMI Admin. 322
BP34
F78533 Buc Cedex, France

Send to your
Country FMI or Service
Administrator

FMI No.: 25343

FMI Title: LightSpeed RT M4 Upgrade

Customer Full Name:

Number and Street Address:

City and Country:

Country Code

Site Number

System I.D.

Unit Tracked:

Model Number	Serial Number	Compl. Date DD/MM/YY	FMI Comp. Code	Service Hours	
				Labor	Travel

FMI Completion Code

1. Modification Complete or previously modified.
2. FMI not required per FMI instruction.
3. Not modified. Unit in storage. FMI kit returned to stock.
4. Modification refused or equipment altered. Customer signature and title required.
5. Unit location unknown or unit not in area. Product Locator notified.

Customer Refusal or Equipment Altered - Modification Pending.

Installer
Comments

Effectivity

All LightSpeed RT systems produce to date (May 10, 2004).

Purpose

The purpose of this FMI is to upgrade all LightSpeed RT sites to an M4 production configuration. The main function of this FMI is to load the M4 version of software on the console. Additionally, there will be a change made to the axial motor windings to achieve more precise alignment positioning capability on the gantry required for Oncology sites.

Furnished Materials

Qty	Description	--FMI Kit 25343, 5113365--	Part #
1	FMI Document		5112555-100
1	GRE Software Installation Procedure		2378261-100
1	CT LightSpeed Series CD		2369740-247
1	LightSpeed 5.X Tips and Workarounds		2407441-100
1	LightSpeed 5.X Technical Reference Manual		2398099-100
1	External Laser Light Disconnect Label		5111615
1	External Laser Label for Cover		5111733
1	LightSpeed Pro & RT Software		5114319
1	P2033YD LSCOM ROM ASSY		2371062-4
1	P2033YA TGP TRIG ROM ASSY		2362522-2
1	Dynamic Brake Fuse		46-170021p106

Tool Required

Qty	Description	Part #
1	LightSpeed 5.X Class C Software	5114786
1	LightSpeed 5.X Class M Software	5114787

People Power

Travel: 1 Hour

Labor: 4 Hours

Procedure

This FMI includes the following procedures:

- Preparing the System for Software Upgrade
- Axial motor winding rewire
- TGP PROM Changes
- Replace Dynamic Brake Fuse
- Load From Cold
- External Laser Characterization Procedure (optional for sites with hospital external lasers)
- Gantry Rotation Characterization Procedure

1 Prepare System for Software Upgrade

1.1 System Operation Verification

Set-up and perform at least one axial scan to verify that the system operated properly before beginning the upgrade

1.2 Software Revision Check

Check the current version of CT Application Software on the system using the showprods command.

1. Open a shell.
2. At the prompt enter the following command
showprods | grep Helios_OC **ENTER**
3. Verify that the response to the showprods inquiry displays one of the following:
I Helios_OC_Product 06/23/2003
LightSpeed ver 501WB.4 rel 501WB.4_H5.2M3

Note: Only LightSpeed 5.X RT Systems currently using the revisions of CT Application Software listed above are compatible with this FMI.

1.3 Axial Motor Winding Change

The LightSpeed RT M4 has changed both the windings of the axial motor and the software to allow for better repeatability of the laser alignment lights. This hardware change along with new software will enable the gantry to “park” more accurately and repeatability. To accommodate this

change in the software, the axial motor windings must be reconfigured to accept this change. This section will walk through the procedure to accomplish this winding change. NOTE: DO NOT SCAN after making this change. This winding change is also dependent upon loading the M4 software on the scanner that will change configuration files to accept the winding change. The scanner will give error messages if a scan is attempted after you have changed the windings

1.3.1 Remove Gantry Side, Bottom, Front and Rear Covers

1.3.2 Power down the gantry and perform Lockout Tagout to insure there are no voltages present during this modification.

1.3.3 Remove motor cover (2 FlatHead Screws)

1.3.4 Change windings from "Low Voltage" to "High Voltage" configuration as shown on the label on the outside cover. The new configuration is shown below. Note: You will need two additional screw caps (wire nuts) to complete this procedure. They are included in the FMI kit

Note: The three larger diameter L1-L3 wires may be labeled as T1-T3. If this is the case, connect these larger diameter wires (T1-T3) to the smaller wire labeled T1-T3



Connect:

T1-L1 T4-T7

T2-L2 T5-T8

T3-L3 T6-T9

Note: Leave P1 and P2 as they

1.3.5 Replace motor cover

1.4 TGP PROM Change

Two PROM devices on the TGP board need to be changed to update the code on this board. The biggest issue this solves is a “trigger jitter” issue, which was present in older versions of these PROM devices. The procedure below details how to change out these devices.

Warning: Please be careful when removing and installing these devices as they can be damaged very easily.

- 1) Verify that the **power to the gantry is still removed.**
- 2) Wear a Wrist Strap and follow proper Anti-Static procedures
- 3) Remove the Sheet Metal Cover on the TGP and MSUB enclosure
- 4) Using a small flathead screw driver or a pair of tweezers, gently pry up I.C. Chips U11 and U18 on the TGP Board (Upper board in this chassis). Pry on the Notched Side (pin 1 end) and then the opposite end switching back and forth a few times to avoid damage to the IC Socket. These Chips are 8 pin DIP Packages and should have small white stickers on them

Note: Be careful and patient while removing these chips, as damage to the circuit board mounted IC socket results in the need to replace the entire circuit board.

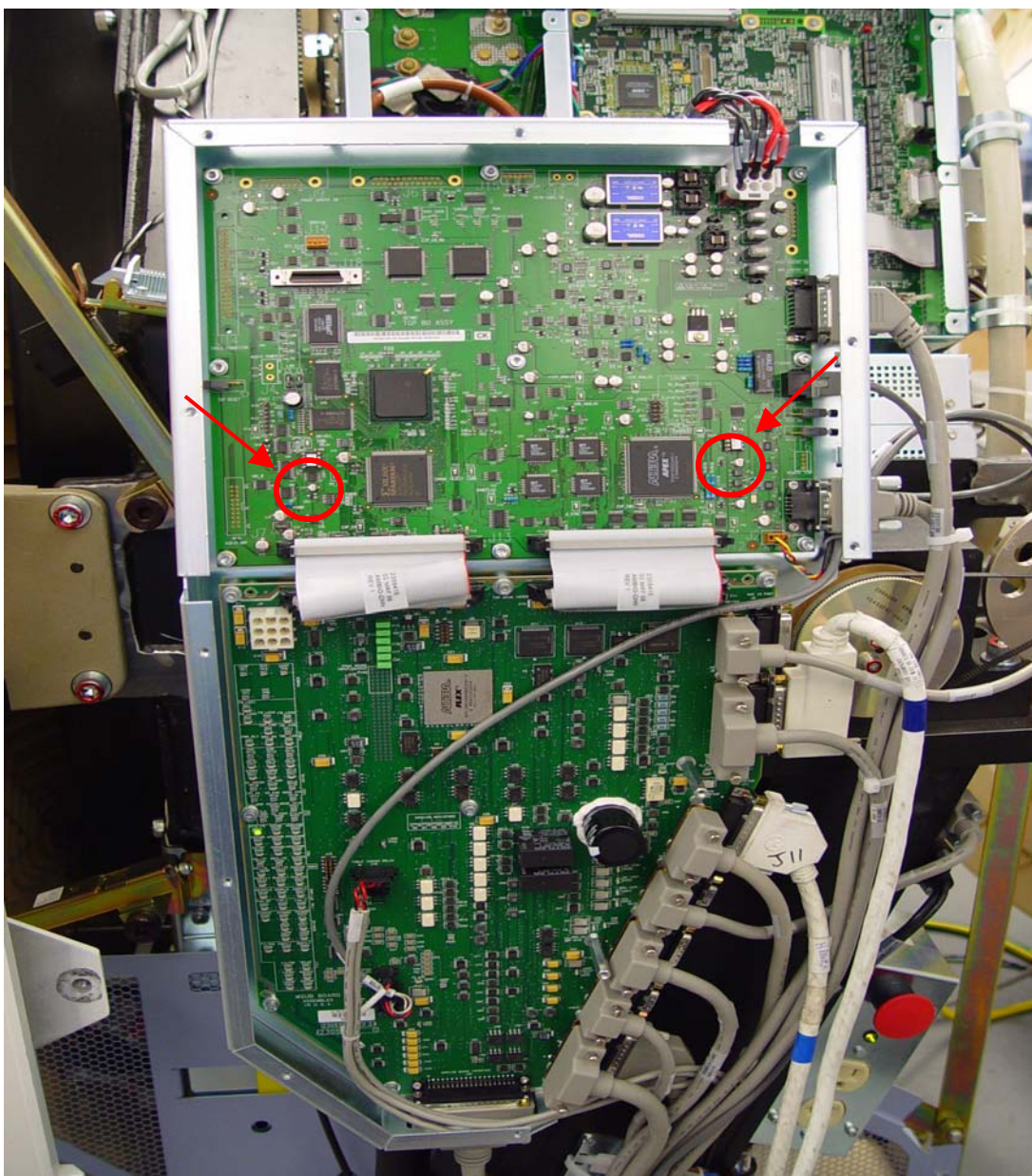


Fig 1: TGP Board on Gantry View

5) Install the replacement chip for U11 in the TGP making sure that the notch on the chip aligns with the notch in the Socket (should be positioned downwards towards the floor). The replacement chip for U11 has a sticker on it labeled 2371062-4. This is the Serial PROM for the Altera Chip on the TGP included in the FMI.

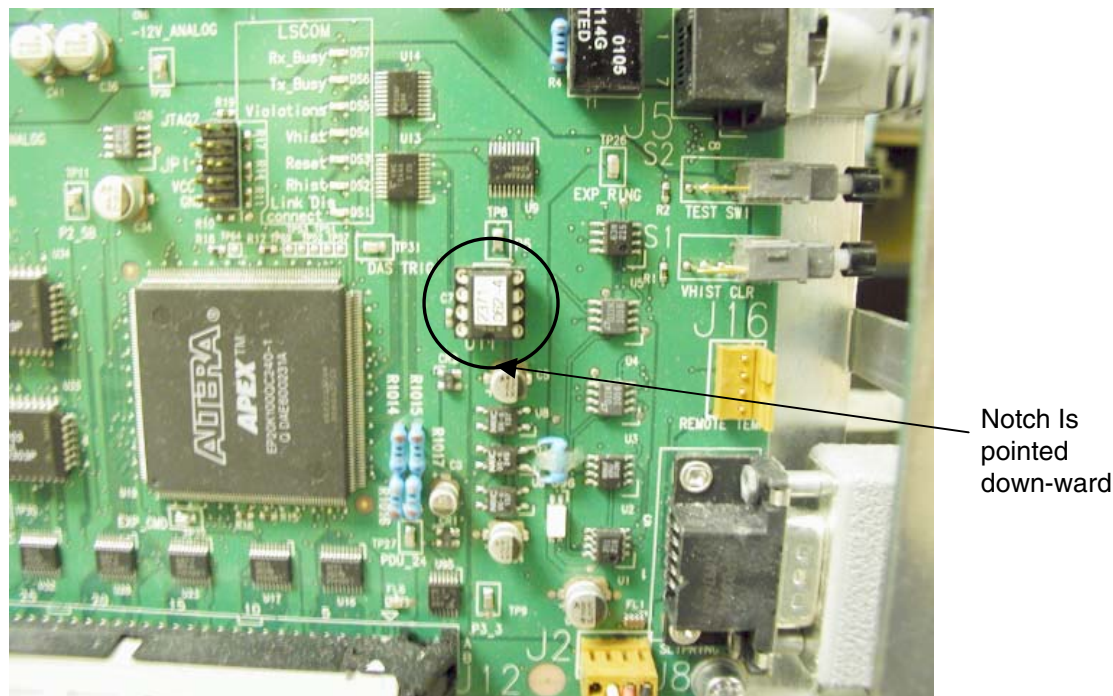


Fig 2: Chip location / installation

6) Install the replacement chip for U18 in the TGP making sure that the notch on the chip aligns with the notch in the socket (should be positioned downwards towards the floor). The replacement chip for U18 has a sticker on it labeled 2362522-5. This is the Serial PROM for the Xilinx Chip on the TGP.

7) Using the Sticker / Label provided change the version of the TGP. The kit provides a small sticker that has the following text on it **2290138-5**. Use this sticker to update the Bar Code Label of the TGP Board. Place the new sticker so it covers up the old board number listed (2290138-04). Position the new sticker so it covers only the old text and leave the Bar Code pattern uncovered.

1.5 Replace Dynamic Brake Fuse

The Axial Dynamic Brake fuse F2 exists to protect the transformer. This fuse can blow due to inrush current for 50Hz systems due to slightly higher operating currents at the lower frequency. The Littlefuse manufactured fuses have a tendency to blow sooner than the Bussman fuses of the same rating. To resolve this issue, the 7A fuse is being replaced with an 8A fuse (46-170021P106 Bussman only). The Hpower Service Fuse kit (2218570-2) is being changed to have the Bussman only 8A fuse (46-170021p106).

Procedure:

1. Verify that the power to the system is still removed.

2. Remove the rear cover.
3. Replace the 7A fuse, shown in figure TBD, below with the 8A fuse supplied in the FMI kit. Using a black marker, relabel near the fuse holder to state "8A".

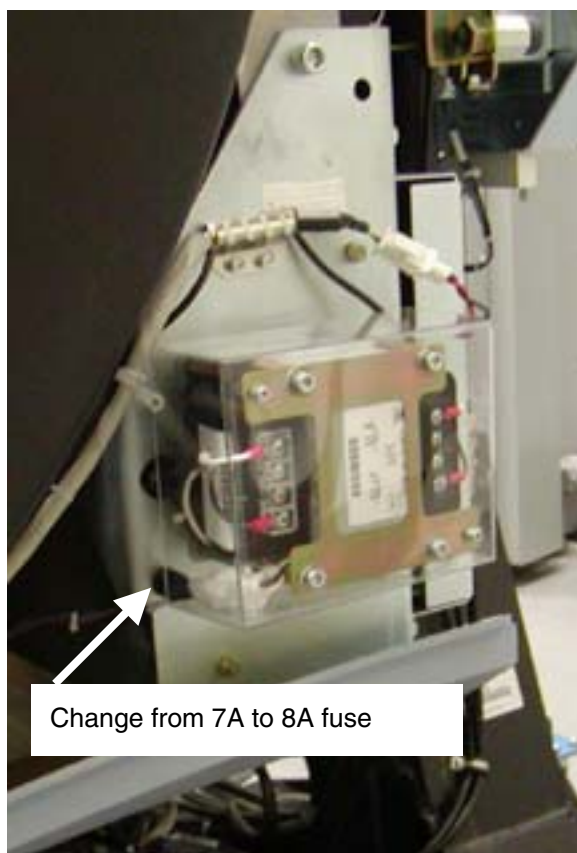


Figure TBD: Dynamic Brake Assembly

4. Restore power to the system.
5. Prescribe a test scan using the service generic protocol 42.1, accept the protocol and scan. If you receive errors and can't scan due to an Axial Drive issue, check the fuse. Disable Axial Drive and reassemble covers. Enable Axial drive prior to putting on the right side cover.

1.6 Archive Image Data

The following procedures (Software Load From Cold) will destroy all image data on the system. Prior to starting the software load, make sure that the customer has reconstructed and archived all images before proceeding.

1.7 Check Software Loading Devices

Since the CDROM Drives on Linux systems are seldom used except for loading system software, it is advisable at this time to perform a quick test on the ability of these devices before you begin the LFC procedure.

Reference: Direction 2378261-100, GRE Software Installation

- Section 3.1.3 Remote Shell – Check Internal Network
- Section 3.1.4 HP Computer CDRom – Mount and Unmount Process
- Section 3.1.5 Peripheral Tower DVD - Mount and Unmount Process
- Section 3.1.6 Peripheral Tower DVD – Initializing a DVD
- Section 3.1.7 Peripheral Tower MOD - Mount and Unmount Process

1.8 Save System State & CSA Protocols

This section details the procedure for archiving the system critical files (System State) and adding to them the customer protocol settings that previously were not included in the System State archive process. This newly created System State will be used during the upgrade Load From Cold Procedure.

Reference: Direction 2378261-100, GRE Software Installation

- Section 3.2.1 Save System State
- Section 3.2.2 CSA Protocol Retention

1.9 Record Key System Configuration Information

To ensure all system configuration information can be restored after the Load From Cold process, record the following parameters. Although the Save System State utility saves most of these parameters, recording this information provides a means to check that these parameters are properly restored after the Restore State Process.

1.10 Network Configuration Info (found in /usr/g/config INFO)

Check CT LightSpeed Network Settings as follows:

1. Open a shell.
2. At the prompt enter the following command

`cd /usr/g/config` **ENTER**

`more INFO` **ENTER**

From the usr/g/config INFO file, record the following network configuration parameters:

Parameter	Line Name in /usr/g/config INFO	Record Parameter Value	Notes
Hospital Name	HOSPITAL_NAME		Can have apostrophe with IIP 3.2
Hostname	GATEWAY_HOSTNAME		
System IP Address	GATEWAY_IP	____.____.____.____	
Broadcast IP Address	GATEWAY_BROADCAST	____.____.____.____	

Parameter	Line Name in /usr/g/config INFO	Record Parameter Value	Notes
Netmask IP Address	GATEWAY_NETMASK	____.____.____.____	
Default Gateway	GATEWAY_DEFAULT	____.____.____.____	May have been used as the Insite Default Gateway in previous BroadBand Insite Checkout.
NIS Server	NIS_SERVER	____.____.____.____	

1.11 HIS/RIS Configuration Info (found in /usr/g/config INFO)

NOTE: If you do not have HIS/RIS, you can skip this section.

Check CT LightSpeed NIS/RIS Settings as follows:

1. Open a shell.
2. At the prompt enter the following command

```
cd /usr/g/config ENTER
```

```
more INFO ENTER
```

From the usr/g/config INFO file, record the following HIS/RIS configuration parameters:

Parameter	Line Name in /usr/g/config INFO	Record Parameter Value	Notes
HIS/RIS Title	HRCTTITLE		
HI/RIS Port Number	HRPORTNUM		
HIS/RIS AE Title	HRAETITLE		
HIS/RIS IP Address	HRIPADDR	____.____.____.____	

1.12 Insite Connectivity Info (see .insiteINFO file)

NOTE: If you don't have Insite, you can skip this section.

1. Open a UNIX shell.
2. At the prompt enter the following command

```
cd /usr/g/insite ENTER
```

```
more .insiteINFO ENTER
```

From the `usr/g/insite .insiteINFO` file, record the following Insite Connectivity configuration parameters:

Parameter	Line Name in <code>/usr/g/insite .insiteINFO</code>	Record Parameter Value	Notes
Connection Type	INS_ConnectionType		
Modem Type	INS_ModemType		Modem Connectivity Only
Modem Phone Number	INS_PhoneNumber		Modem Connectivity Only
Modem Dial Prefix	INS_DialPrefix		Modem Connectivity Only
Modem Tone or Pulse Setting	INS_ToneOrPulse		Modem Connectivity Only
Modem Country	INS_Country		Modem Connectivity Only
Modem String	INS_ModemString		Modem Connectivity Only
Modem TTY Port	INS_TtyPort		Modem Connectivity Only
Modem TTY Speed	INS_TtySpeed		Modem Connectivity Only
BroadBand Insite Gateway IP	INS_GatewayIP	---.---.---.---	BroadBand Connectivity Only
BroadBand Gateway Mask	INS_GatewayMASK	---.---.---.---	BroadBand Connectivity Only

1.13 BroadBand Insite Connected Systems

If your system has already been Insite Checked Out as a Broadband system, you will need to re-enter the Insite Gateway IP Address into the IIP Admin Configuration utility when reconfiguring Insite after the Load From Cold process.

To ensure that you enter the correct IP Address of the Insite dedicated Gateway, confirm this IP Address with one of the following resources:

- Ask the Hospital Network Administrator or IT contact.
- In the United States, check for the IP Address at the following web site or by calling the regular checkout number (800-321-7937):
<http://gein2.med.ge.com/vpni/secure/index.jsp>
- In Europe, contact the regular toll free number to reach the OLCE.

IP Address: _____

Perform this investigation prior to arriving on site.

1.14 Installed Options (found using swokinstall -p)

Check CT LightSpeed RT Options Settings as follows:

- 1) Open a UNIX shell.
- 2) At the prompt enter the following command

swokinstall -p **ENTER**

From the swokinstall -p output, record which of the following LightSpeed 16 system options are installed on the system:

Line Name in output file of swokinstall -p	Enabled (Y/N)	Permanent	Flex Trial	Standard LightSpeed 16 Option or Purchased Option
Smart Prep				Standard
Wide View				Standard
Helical Tilt				Standard
Direct-3D				Standard
VariViewer				Standard
AutomA				Standard
Power440				Standard
Large Image Series				Standard
Thin Twin Helical				Standard
Connect Pro				Purchased Option
DentaScan				Purchased Option
Smart Score				Purchased Option
Smart Score Pro				Purchased Option
CardIQ SnapShot				Purchased Option
Smart Step				Purchased Option
CTPerfusion2				Purchased Option
VolumeViewer				Purchased Option
AdvVesselAnalysis				Purchased Option
CTColonoscopy				Purchased Option
CT Perfusion3				Purchased Option
Advantage 4D				Purchase Option
Auto Bone				Purchased Option

NOTE: Make sure that for each Option enabled, there is an on-site DVD that contains the key(s) for the Options presently enabled. Without the DVD keys, you will not be able to restore options on the system that were previously enabled.

2 Load From Cold Procedure

After system operation has been verified, all images have been archived, a Save System State has been generated, and a record of key system configuration parameter has been completed, you are ready to upgrade the system software.

2.1 Shutdown Through Start CT Applications Steps

Linux/GRE Consoles

Using the LightSpeed GRE Software Installation Procedure document (2378261-100) and the software included in the FMI kit, perform a software load.

Reference: Direction 2378261-100 revision 10, GRE Software Installation. This is included in the FMI.

2.2 Restore System To Operational Conditions

Linux/GRE Consoles

Using the LightSpeed GRE Software Installation Procedure document (2378261-100) and the software included in the FMI kit, perform a software load.

Reference: Direction 2378261-100, GRE Software Installation

Section 3.8	Restore System State (and CSA Protocols)
Section 3.10	Retro Recon Test
Section 3.11	FLASH Download

NOTE: Make sure that the Key System Information recorded in Section 1 (Network Configuration Info and HIS/RIS Configuration Info) is correctly set as recorded. If not, use the following to:

- reconfig utility to reset Network Configuration Info.
- Connect Pro utility to reset the HIS/RIS Configuration Info.

2.3 Restore System Options

Using the record of installed options prior to the software load, install all the options using the DVD keys for each.

Reference: Direction 2378261-100, GRE Software Installation

Section 3.12	Install Options
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2.4 Complete the Load From Cold Process

Reference: Direction 2378261-100, GRE Software Installation

Section 3.6	Configure Users System Access
Section 3.7	HIPAA Present set to ON – Admin Screen
Section 3.13	Set Time and Date
Section 3.14	Restart the System
Section 3.15	System Sanity Scanning

NOTE: AVOID LOSING INSITE CONNECTIVITY CONFIGURATION DATA!!

DO NOT SAVE SYSTEM STATE UNTIL YOU'VE COMPLETED LOADING CLASS C AND CLASS M SOFTWARE AND RE-INITIALIZED INSITE CONNECTIVITY. PERFORMING THE SAVE STATE PROCESS PRIOR TO LOADING CLASS C AND CLASS M SOFTWARE MAY CAUSE THE LOSS OF CRITICAL INSITE CONFIGURATION PARAMETERS AND RESULT IN THE NEED TO PERFORM AN INSITE CHECKOUT TO RESTORE CONNECTIVITY.

4 Load Class C and Class M Software

Notice: Avoid Partial Load of Class M Software

To ensure a proper load of the Class M Software, load Class C Software and immediately load Class M Software without rebooting the system between CD loads. The Class M Software CD load will fail to mount if not completed in this sequence. If the Class M Software install log exhibits Errors, usually a repeat attempt at loading Class M Software CD succeeds.

4.1 Load Class C Software

Load provided Class C software as follows:

1. Open a shell.
2. Enter root as superuser:
 - a. At the ctuser prompt, type: `su -` **ENTER**
 - b. At the Password prompt, type: `#bigguy` **ENTER**
3. Insert the Class C software CD into HP computer DVD Drive. Wait for the disk read to STOP.
4. Install the Class C software on the system. At the root prompt, type:
`install.classC` **ENTER**
5. Close Class C install reporting shell.
6. Verify Class C is properly loaded on the system.
 - a. At the root prompt, type: `cd /var/adm` **ENTER**

- b. At the root prompt, type: `ls` **ENTER**
 - c. Find the `install.classc.log` file. If file found, this means the Class C software load completed.
 - d. At the root prompt, type:
`more install.classc.log.<yyyymmddhhmmss>`
Refer to Appendix A for an example of a successful Class C software load.
7. Remove the Class C software CD from HP computer DVD Drive.

4.2 Load Class M Software

Load provided Class M software as follows:

1. Open a shell.
2. Enter root as superuser:
 - a. At the ctuser prompt, type: `su -` **ENTER**
 - b. At the Password prompt, type: `#bigguy` **ENTER**
3. Insert the Class M software CD into the HP computer DVD drive. Wait for the disk read to STOP.
4. Install the Class M software on the system. At the root prompt, type:
`install.classM` **ENTER**
5. Verify Class M is properly loaded on the system.
 - a. At the root prompt, type: `cd /var/adm` **ENTER**
 - b. At the root prompt, type: `ls` **ENTER**
 - c. Find the `install.classM.log` file. If file found, this means the Class M software load completed.
 - d. At the root prompt, type:
`more install.classM.log.<yyyymmddhhmmss>`
Refer to Appendix A for an example of a successful Class M software load.
6. Remove the Class M software CD from HP computer DVD Drive.

5 Re-Initializing Insite Connectivity

This release of software contains a new IIP release (IIP 3.2-Linux). Changes in this IIP release are primarily in the following areas:

- Updates to support Multi-Tech modem firmware revision 8.19M
- Removal of IIP_DialPrefix ProDiags task
- Replacement of current iipHealthPage ProDiags task with a new Insite Data Mining task
- Auto Route to AutoSC in Level 5 setup

In addition, the Network Configuration Tab in the Insite Interactive Administration utility has been modified to eliminate the Use Default Gateway selection.

To ensure these IIP 3.2 features are correctly set, an Insite Checkout is NOT required. Because the basic Insite configuration information on the system should be correct from the original Insite Checkout, in most cases, it is only necessary to run ConfigLink via exiting the IIP Admin Configuration utility Insite Checkout Tab Screen.

5.1 Restore System State To Retrieve Class M Configuration Parameters

Once Class M software has been loaded, you must repeat a Restore System State to recover all previous Insite and Class M configuration parameters.

Reference: Direction 2378261-100, GRE Software Installation

Section 3.9 Restore System State

NOTE: Make sure that the Key System Information recorded in Section 1 (Insite Connectivity Info) is correctly set as recorded. If not, use the IIP Admin Configuration utility to reset Network Configuration Info as described in the following procedures.

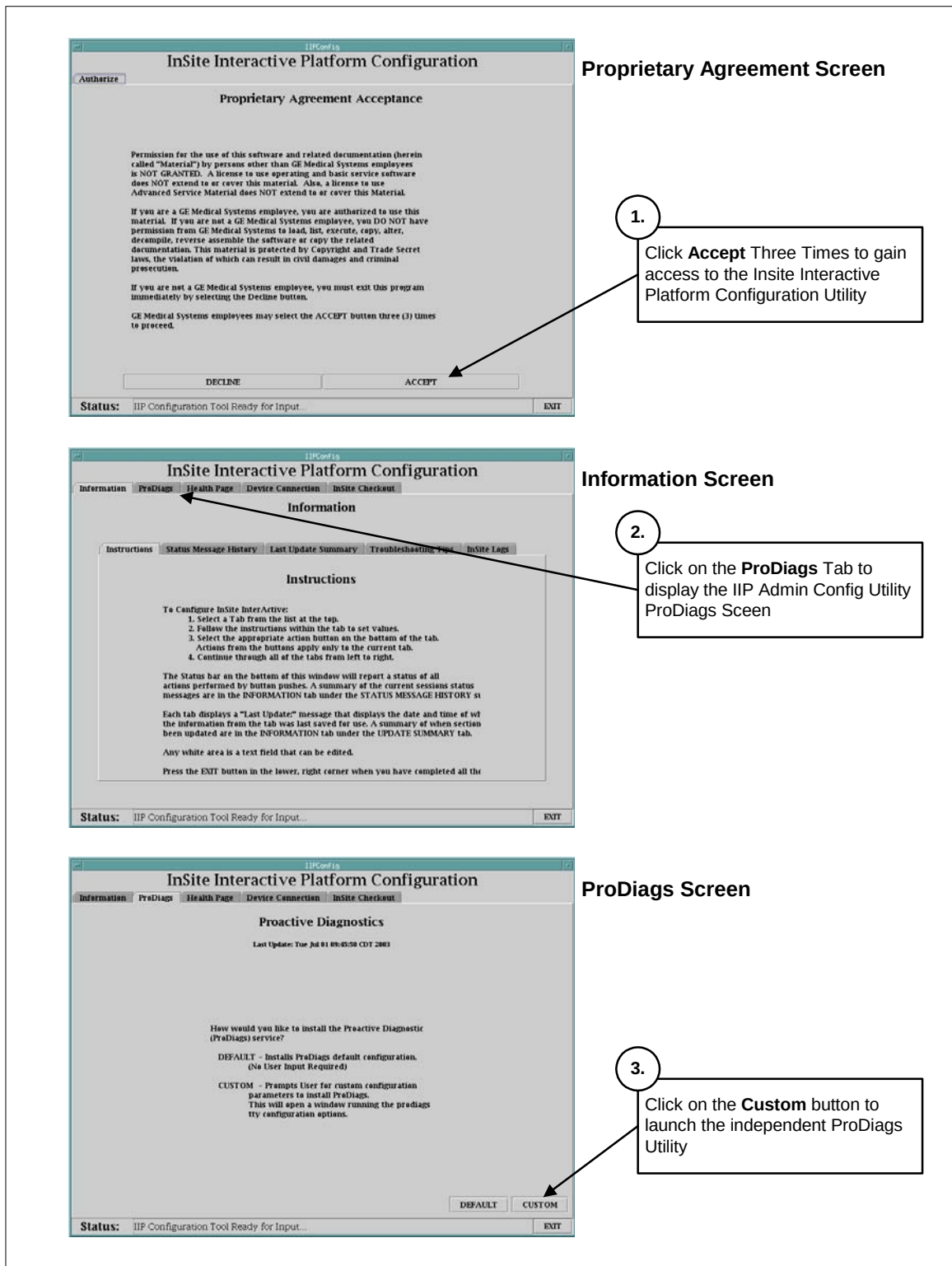
5.2 Enable Insite Interactive Platform Configuration Utility

See Figure 2, Insite Interactive Platform Configuration Utility Gaining Access and Selecting ProDiags Utility.

Enable the Insite Interactive Platform Configuration utility as follows:

1. Open a shell.
2. Enter root as superuser:
 - a. At the ctuser prompt, type: `su -` **ENTER**
 - b. At the Password prompt, type: `#bigguy` **ENTER**
3. Launch the IIP Admin Configuration utility. At the root prompt, type:
`iipadmin config` **ENTER**
The system displays the Insite Interactive Platform Configuration utility with the Proprietary Agreement Acceptance Screen.
4. Click on **ACCEPT** three times to display the Information Screen.
5. Click on the **ProDiags** Tab to display the ProDiags Screen.
6. Click on the **Custom** button to launch the independent ProDiags utility as shown in Figure 3.

Figure 2
Insite Interactive Platform Configuration Utility
Gaining Access and Selecting ProDiags Utility



5.3 Configure ProDiags Schedule

With this release of software, the IIP 3.2 package contains a new data collection package that pushes gesyslog data and a basic system configuration file to the On Line Center RDA Database. This IDM package contains two new ProDiags tasks (idm and idm_send_data) that must be configured to match the site's operational usage.

The default ProDiags schedule provided in this release should be adequate for most systems. However, we recommend that you review this schedule and update according to your particular site's normal operation.

Updating ProDiags Scheduled Tasks

See Figure 3 and refer to detailed steps on the following page.

Figure 3
ProDiags Graphical Utility
Schedule Tab – Modifying Scheduled Tasks

Proactive Diagnostics

Schedule Management

Schedule Management provides the following functionalities:

1. Add a task to the schedule
2. Delete a task from the schedule
3. View the schedule

Task	Iterations	Day	Hour	Minute	Type	Time Slot
PassChg	1	Daily	05 AM	30	Background	
pd_HouseK...	1	Daily	04 AM	30	Background	
pd_ASC No...	1	Daily	Hourly	28	Background	
health...	1	Daily	04 AM	45	Background	
IIP_HouseK...	1	Daily	06 AM	12	Background	
IIP_HouseK...	1	Daily	12 PM	12	Background	
IIP_HouseK...	1	Daily	06 PM	12	Background	
IIP_HouseK...	1	Daily	12 AM	12	Background	
idm_send...	1	Daily	07 AM	00	Background	
idm	1	Daily	Hourly	00	Background	

Buttons: New, Delete, Reload, Apply, Help

Status: Reading data... Done

Callouts:

1. Click on **Hour** and **Minute** fields of idm_send_data task and select an hour and minute from the pull down lists between 9 AM and 5 PM. After each selection, click **Yes** on the "Data has been modified. Please confirm it" pop up.
2. Click on **Hour** and **Minute** fields of idm task and select an hour and minute from the pull down lists to schedule 5 minutes before the idm_send_data task. After each selection, click **Yes** on the "Data has been modified. Please confirm it" pop up.
3. When idm and idm_send_data Hour and Minute selections are complete, click on **Apply** to finalize the schedule updates. Click **Yes** on the "Are you sure you want to apply changes?" pop up.
4. Close the ProDiags utility browser by clicking on **Close** from the upper left browser window pull down menu.

Note: If

Use the following sequence to update the scheduled time (Hour and Minute) of the `idm` and `idm_send_data` tasks:

ProDiags `idm_send_data` task

- Click on the **`idm_send_data`** task **Hour** field.
- From the pull down list, select an hour that the system is normally on (typically between 09:00 and 17:00).
- On the “Data has been modified. Please confirm it” pop-up, click on **YES**.
- Click on the **`idm_send_data`** task **Minute** field.
- From the pull down list, select a minute (preferably an odd minute such as 3, 14, 22, etc).
- On the “Data has been modified. Please confirm it” pop-up, click on **YES**.

ProDiags `idm` task: Set up the `idm` task to run five minutes before the `idm_send_data` task.

- Click on the **`idm`** task **Hour** field.
- From the pull down list, select an hour that sets the `idm` task to run five minutes before the `idm_send_data` task.
- On the “Data has been modified. Please confirm it” pop-up, click on **YES**.
- Click on the **`idm_send_data`** task **Hour** field.
- From the pull down list, select a minute that sets the `idm` task to run five minutes before the `idm_send_data` task.
- On the “Data has been modified. Please confirm it” pop-up, click on **YES**.

When completed with the `idm` and `idm_send_data` modifications, click on the **APPLY** button at the bottom of the the Schedule Tab. On the “Apply change confirmation” pop-up, click on **YES**.

NOTICE: If the schedule change failed due to a permissions denied error, execute the following change-mode command to make the file named `bg_schedule_data` read/writable:

- With the ProDiags GUI still open, open another shell
- Become root user:
At the ctuser prompt, type: `su -` **ENTER**
At the Password prompt, type: `#bigguy` **ENTER**
- Set Permissions for `bg_schedule_data`:
type: `cd /usr/g/insite/ProDiags.schedule`
type: `chmod 777 bg_schedule_data`

Repeat the Apply click on the ProDiags Schedule Tab. The update should occur without failures.

Close the ProDiags utility browser by selecting **Close** from the upper left browser window pull down menu. The ProDiags browser should close and the Insite Interactive Platform Configuration utility should display the HealthPage Tab as shown in Figure 4.

5.4 Configure HealthPage

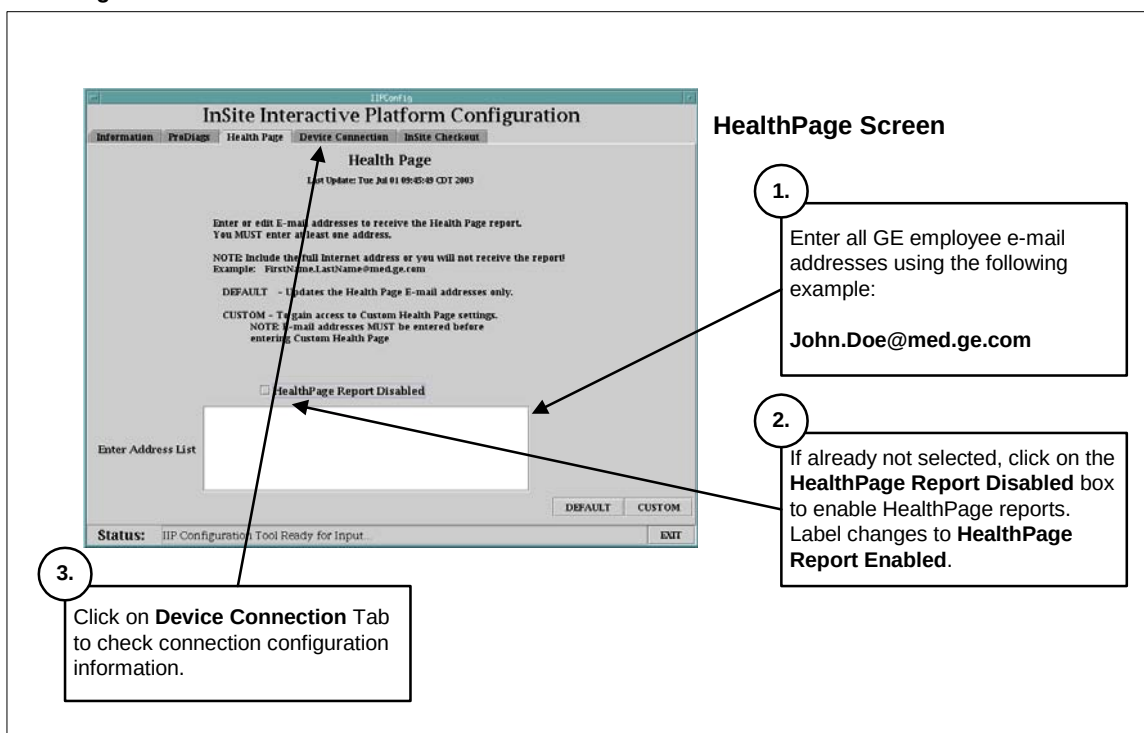
After completing the ProDiags setup, the IIP Admin Configuration utility automatically switches to the HealthPage Tab as shown in Figure 4.

The Restore State process (after loading Class M software) restored the following information for the HealthPage Reports:

- All e-mail addresses of targeted recipients that were previously loaded
- HealthPage report contents that was previously configured
- HealthPage report schedule that was previously configured

Unless you have recent changes to any of these three items, you can skip the HealthPage Tab of the IIP Admin Configuration utility and select the Device Connection Tab. If you have changes that you want to make (e-mail addresses, report content, report schedule), follow the instructions on the IIP Admin Configuration utility.

Figure 4
Insite Interactive Platform Configuration Utility
HealthPage Tab



5.5 Check Device Connection Setup

5.5.1 Modem Setup

When selecting the Device Connection Tab, the IIP Admin Configuration utility automatically displays the Modem Setup Screen as shown in Figure 5.

If your system was configured for Modem connectivity, the Restore State process (after loading Class M software) restored all the Modem setup parameters. To verify, refer to Section 1 of this FMI where you recorded Insite Connectivity Info.

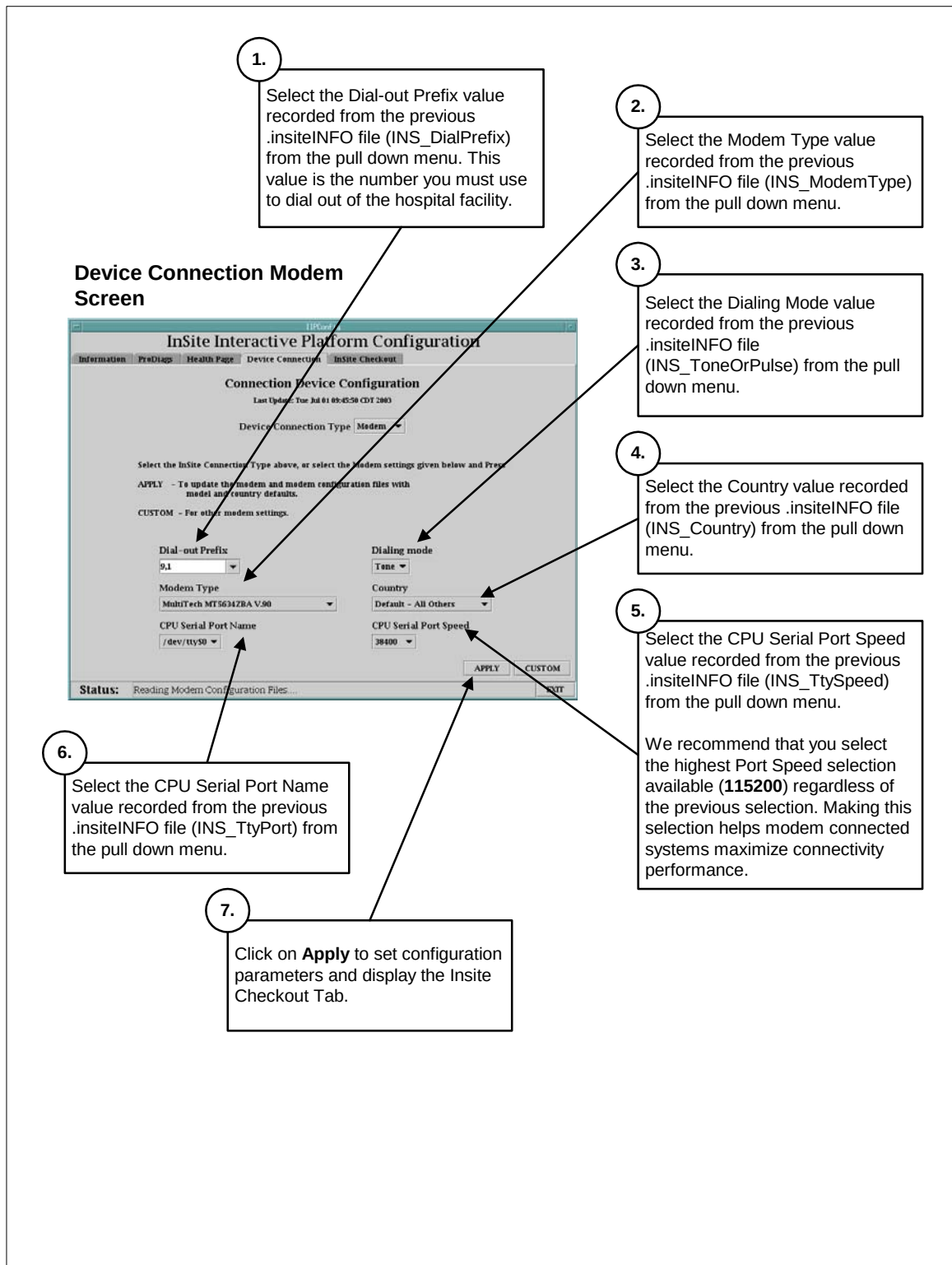
- If there were no Modem parameters in the .insiteINFO file but there were BroadBand parameters, skip to the Check BroadBand Device Connection Setup procedure.
- If there were Modem parameters in the .insiteINFO file but the parameters displayed in the Modem Setup Screen do not match what you previously recorded, enter the correct values as required.

NOTE: The Modem configuration parameters should be pre-loaded from the Restore State process. Do not change settings unless these values were not properly restored.

If required, follow the instructions in Figure 5. The default values are usually the correct values or most commonly used. Parameter selections that do not require customization are:

CPU Serial Port Name

Figure 5
Insite Interactive Platform Configuration Utility
Device Connection Tab – Modem Setup



5.5.2 BroadBand Setup

If your system was previously BroadBand Insite connected, click on the Network button of the IIP Admin Configuration utility Device Connection Tab to display the Network Setup Screen as shown in Figure 6.

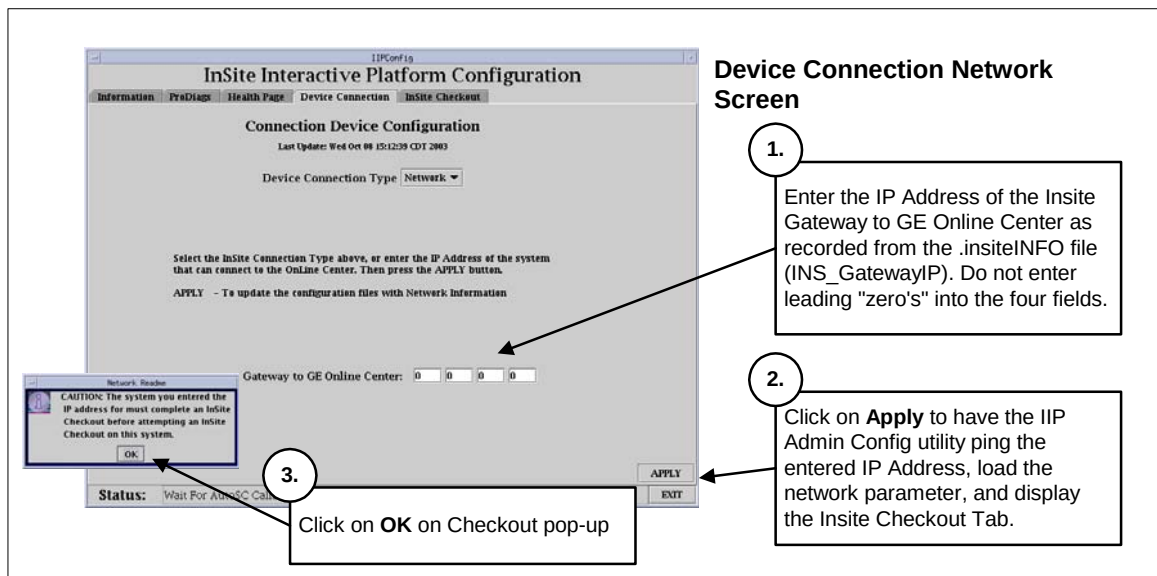
If your system was configured for BroadBand connectivity, the Restore State process (after loading Class M software) restored all the BroadBand setup parameters. To verify, refer to Section 1 of this FMI where you recorded Insite Connectivity Info.

NOTE: The BroadBand Network configuration parameters should be pre-loaded from the Restore State process. However, the on-site FE who performed the original Insite Checkout of the system may have used the hospital's imaging LAN default gateway IP Address (selection was available in the previous release of IIP).

DO NOT USE THE HOSPITAL WORKING LAN DEFAULT GATEWAY IP ADDRESS AS THE INSITE GATEWAY IP ADDRESS OTHER THAN AS A TEMPORARY MEANS TO CONNECT TO THE ON LINE CENTER!!

To find the correct Insite Gateway IP Address, use the contacts listed in Section 1.10, BroadBand Insite Connected Systems.

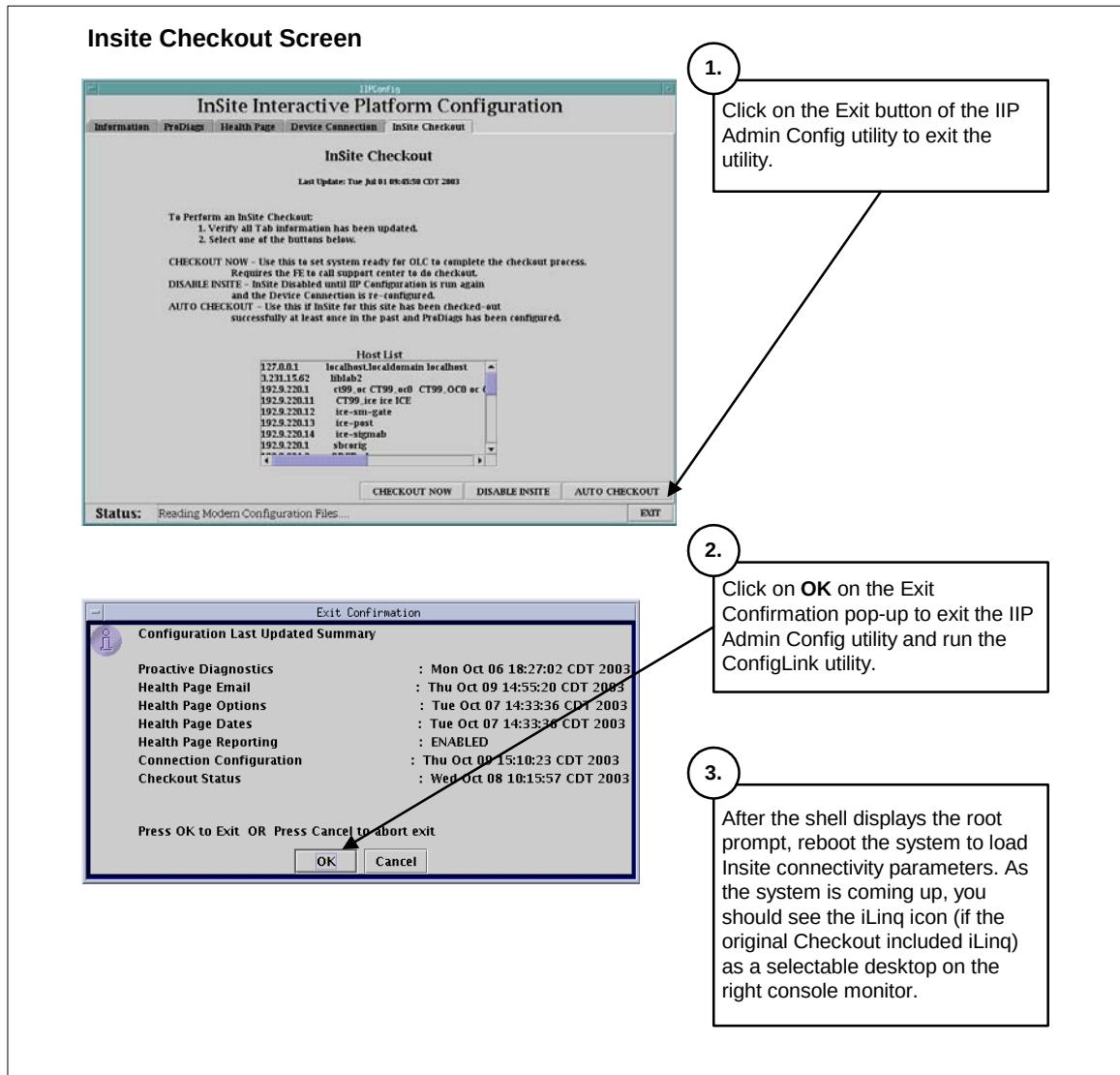
Figure 6
Insite Interactive Platform Configuration Utility
Device Connection Tab – Network Setup



5.6 Completing the Insite Re-Initialization

Follow the instructions provided in Figure 7 to exit the IIP Admin Configuration Utility.

Figure 7
Insite Interactive Platform Configuration Utility
Insite Checkout Tab – Exiting the IIP Admin Configuration Utility



5.7 Run ConfigLink

Run the Insite **ConfigLink** script to populate all configuration information required to fully restore connectivity parameters on the system.

1. Open a shell.
2. Enter root as superuser:
 - a. At the ctuser prompt, type: **su -** **ENTER**
 - b. At the Password prompt, type: **#bigguy** **ENTER**
3. Change directory to Insite. At the root prompt, type:
cd /usr/g/insite **ENTER**
4. Run the ConfigLink script. At the root prompt, type:
ConfigLink **ENTER**

Wait for the shell to return the root prompt to ensure the **ConfigLink** process completes. When complete, exit root and close the shell.

5.8 Reboot System

To ensure all Insite configuration parameters are properly loaded into the operating system (route tables, etc.), perform a system reboot.

1. Click on the pink Shutdown button.
2. From the Shutdown selection pop-up, click on **RESTART**, then **OK**, to reboot the system.

5.9 Save System State

At the completion of the Load From Cold Process and Insite Re-Initialization process, ensure all system state parameters are saved using the Save System State Utility.

Reference: Direction 2378261-100, GRE Software Installation

Section 3.2.1 Save System State

Reference: Direction 236590-100, Linux/Pegasus Software Installation

Section 2.3 Save System State

6 External Alignment Light Characterization

The external alignment characterization procedure is a new feature for the LightSpeed RT M4. If the hospital site has external lasers mounted in the exam room, this feature gives the ability to characterize the distance from the internal lasers to the hospital supplied external lasers instead of the gantry external lasers. This procedure is found on the LightSpeed 5.X service CD (Part Number 2373055-200). The alignment light characterization procedure is under the calibration tab on this CD.

Please note that if the hospital does not have these external alignment lights, this procedure does not have to be completed.

The stickers referenced by the service procedure as found in the install support kit are supplied in this FMI instead. They will be supplied with the install support kit for all future system shipments.

7 Gantry Characterization Procedure

The gantry rotation characterization procedure is a new feature for the LightSpeed RT M4. Clinical feedback from the pilot sites has found that images on a LightSpeed RT scanner can be rotated when viewed from the console due to potential errors in gantry home location. This is assuming the "C-Pulse alignment" for the gantry home flag has been properly aligned. As a result, a new characterization procedure was developed to back characterize this rotation offset error into the firmware and effectively level the image on the console. Please reference the LightSpeed 5.X service CD (Part Number 2373055-200). The gantry characterization procedure is under the calibration tab on this CD.

8 Finishing Up

8.1 Customer Documentation & Software

Make sure that the customer receives the Class A Operator/Service documentation and all the Software CD-ROM's provided with this FMI.

8.2 FMI Closure

Close FMI per regional procedure. Note that the FMI cannot be closed until the Product Locator cards are completed, sent in, and GIB is updated with all tracked items in this FMI.

Remove this FMI document from the customer site.

Appendix

Enable CT Number Range (RT Only)

For Enable CT Number Range (RT Only) please refer to LIGHTSPEED 5.X INSTALLATION MANUAL 2349182-100 (page 162 for Revision 6).